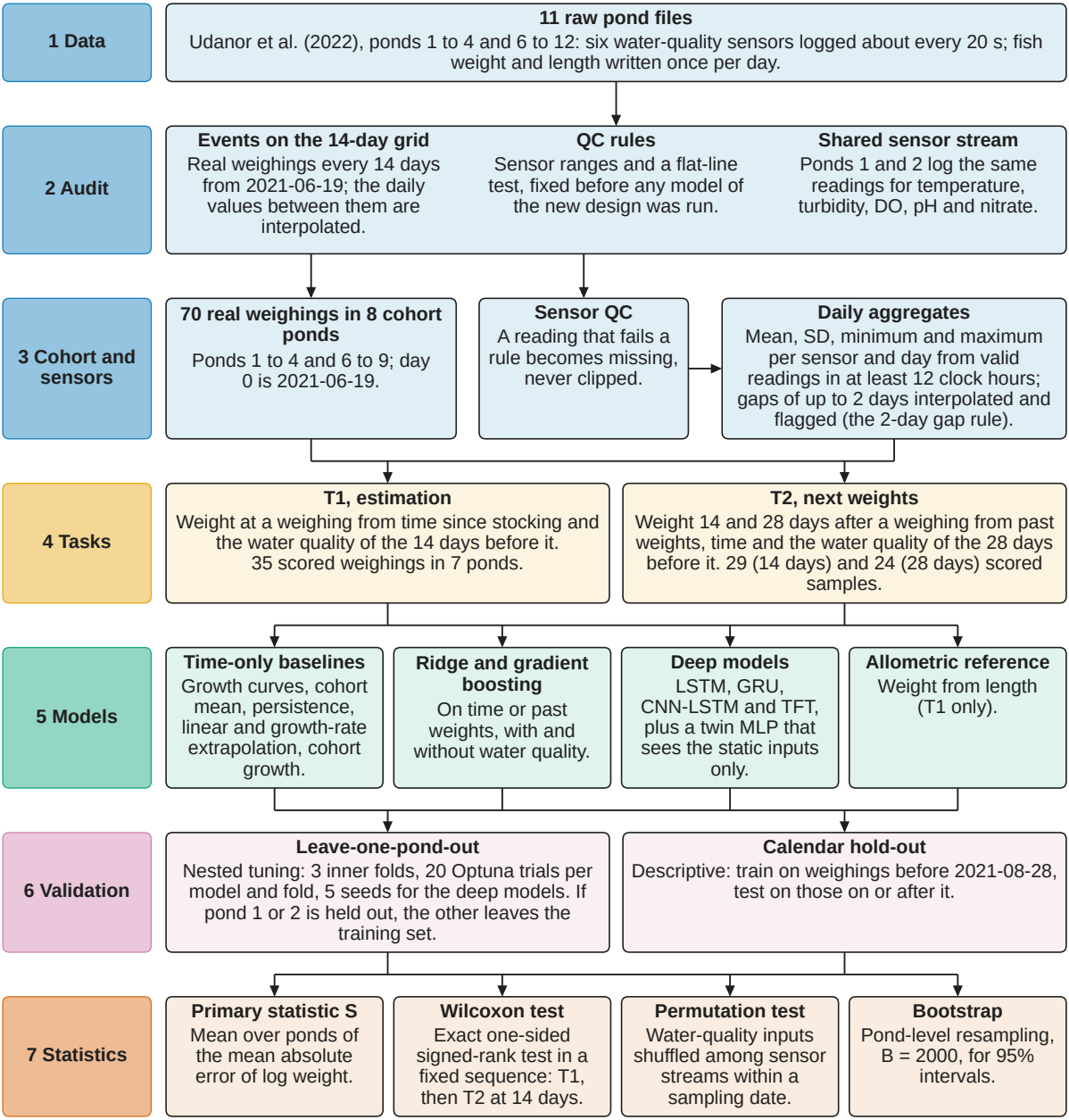
1 Data
11 raw pond files
Udanor et al. (2022), ponds 1 to 4 and 6 to 12: six water-quality sensors logged about every 20 s; fish weight and length written once per day.
2 Audit
Events on the 14-day grid
Real weighings every 14 days from 2021-06-19; the daily values between them are interpolated.
QC rules
Sensor ranges and a flat-line test, fixed before any model of the new design was run.
Shared sensor stream
Ponds 1 and 2 log the same readings for temperature, turbidity, DO, pH and nitrate.
3 Cohort and sensors
70 real weighings in 8 cohort ponds
Ponds 1 to 4 and 6 to 9; day 0 is 2021-06-19.
Sensor QC
A reading that fails a rule becomes missing, never clipped.
Daily aggregates
Mean, SD, minimum and maximum per sensor and day from valid readings in at least 12 clock hours; gaps of up to 2 days interpolated and flagged (the 2-day gap rule).
4 Tasks
T1, estimation
Weight at a weighing from time since stocking and the water quality of the 14 days before it. 35 scored weighings in 7 ponds.
T2, next weights
Weight 14 and 28 days after a weighing from past weights, time and the water quality of the 28 days before it. 29 (14 days) and 24 (28 days) scored samples.
5 Models
Time-only baselines
Growth curves, cohort mean, persistence, linear and growth-rate extrapolation, cohort growth.
Ridge and gradient boosting
On time or past weights, with and without water quality.
Deep models
LSTM, GRU, CNN-LSTM and TFT, plus a twin MLP that sees the static inputs only.
Allometric reference
Weight from length (T1 only).
6 Validation
Leave-one-pond-out
Nested tuning: 3 inner folds, 20 Optuna trials per model and fold, 5 seeds for the deep models. If pond 1 or 2 is held out, the other leaves the training set.
Calendar hold-out
Descriptive: train on weighings before 2021-08-28, test on those on or after it.
7 Statistics
Primary statistic S
Mean over ponds of the mean absolute error of log weight.
Wilcoxon test
Exact one-sided signed-rank test in a fixed sequence: T1, then T2 at 14 days.
Permutation test
Water-quality inputs shuffled among sensor streams within a sampling date.
Bootstrap
Pond-level resampling, B = 2000, for 95% intervals.